Circle
Part 2

4) Equation of tangent at (x_1, y_1) to the circle $x^2 + y^2 + 2mx + 2ny + c = 0$

The center of the circle is $(-m, -n)$. The slope of the radius on the tangent is $\frac{y_1+n}{x_1+m}$

Hence the equation of the tangent is $\frac{y-y_1}{x-x_1} = -\frac{x_1+m}{y_1+n}$.

$$\therefore (y - y_1)(y_1 + n) + (x - x_1)(x_1 + m) = 0$$

$$yy_1 + ny - {y_1}^2 - ny_1 + xx_1 + mx - {x_1}^2 - mx_1 = 0$$

$$xx_1 + yy_1 + mx + ny = {x_1}^2 + {y_1}^2 + mx_1 + ny_1$$

Adding $mx_1 + ny_1$ *to both sides*

$$xx_1 + yy_1 + m(x + x_1) + n(y + y_1) = {x_1}^2 + {y_1}^2 + 2mx_1 + 2ny_1$$
$$= -c$$

Where the point (x_1, y_1) satisfies the equation of the circle

$\therefore$ The equation of the tangent at point (x_1, y_1) is

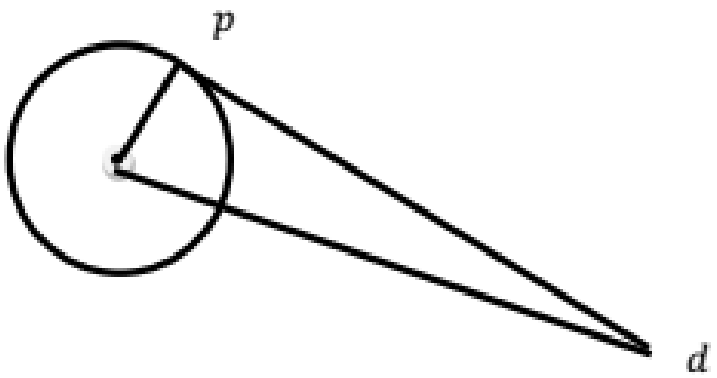
$$\boxed{xx_1 + yy_1 + m(x + x_1) + n(y + y_1) + c = 0}$$

5) Length of tangent from point $p(x_1 , y_1)$ to the circle $x^2 + y^2 + 2mx + 2ny + c = 0$

Since the center of the circle is $(-m, -n)$ and

the radius is

$r = \sqrt{m^2 + n^2 - c}$.



From the figure, we find that

$\overline{pd}^2 = \overline{od}^2 - \overline{op}^2$

$= (x_1 + m)^2 + (y_1 + n)^2 - r^2$

$= {x_1}^2 + 2m\, x_1 + m^2 + {y_1}^2 + 2n\, y_1 + n^2 - m^2 - n^2 + c$

$= {x_1}^2 + {y_1}^2 + 2mx_1 + 2ny_1 + c$

$$\boxed{pd = \sqrt{{x_1}^2 + {y_1}^2 + 2mx_1 + 2ny_1 + c}}$$

6) The condition of a straight line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$.

The center of the circle is $(0, 0)$. The radius of the circle is a. If $y = mx + c$ is a tangent to the circle, the perpendicular distance from the centre on the straight line $y = mx + c$ is the radius of the circle.

$$\pm a = \frac{c}{\sqrt{1+m^2}} \qquad c^2 = a^2(1 + m^2)$$

Solved Problems

Example

Find the equation of the circle that touches the line $x + y - 1 = 0$ and passes through the points (2,1), (0,5), then find its center and radius.

Solution

Suppose that the equation of the circle is

$$x^2 + y^2 + 2mx + 2ny + c = 0$$

$\because$ the circle passes through the points (2,1), (0,5)

$\therefore$ it verifies the equation of the circle, that is,

$at \quad (0,5) \quad \Rightarrow \quad 0 + 25 + 10n + c = 0$

$10\,n + c = -25 \qquad (1)$

$at\ (2,1) \quad \Rightarrow \quad 4 + 1 + 4m + 2n + c = 0$

$4m + 2n + c = -5 \qquad (2)$

$\because$ the circle touches the line $x + y - 1 = 0$

$\therefore$ The length of perpendicular straight line from the center $(-m, -n)$ to the line $x + y - 1 = 0$ is the radius

$$r = \sqrt{m^2 + n^2 - c} = \left|\frac{-m - n - 1}{\sqrt{1+1}}\right| = \frac{m + n + 1}{\sqrt{2}}$$

$$\therefore m^2 + n^2 - c = \frac{(m + n + 1)^2}{2}$$

$$m^2 - 2nm + n^2 - 2m - 2n - 2c = 1 \qquad (3)$$

By solving equations (1), (2), (3), we find that

$$m = -19 - 8\sqrt{5}, n = 4\left(-3 - \sqrt{5}\right), c = 5\left(19 + 8\sqrt{5}\right)$$

Substituting in the equation of the circle for m, n, c, we get the equation of the circle.

Example

Prove that the line $x + y + 2 = 0$ touches the circle $x^2 + y^2 + x + y = 0$. Then find the point of contact.

Solution

The center of the circle is $\left(\frac{-1}{2}, \frac{-1}{2}\right)$ and the radius is $r = \sqrt{\frac{1}{4} + \frac{1}{4} - 0} = \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$

The length of perpendicular straight line from $\left(\frac{-1}{2}, \frac{-1}{2}\right)$ to $x + y + 2 = 0$ is

$$d = \left|\frac{ax_1 + b\,y_1 + c}{\sqrt{a^2 + b^2}}\right| = \left|\frac{\frac{-1}{2} - \frac{1}{2} + 2}{\sqrt{1+1}}\right| = \left|\frac{1}{\sqrt{2}}\right| = \frac{1}{\sqrt{2}}$$

Since $r = d = \frac{1}{\sqrt{2}}$, then the line $x + y + 2 = 0$ touches the circle $x^2 + y^2 + x + y = 0$.

To find the point of contact, we solve the equation of the line with the circle

$$x^2 + (x + 2)^2 + x - (x + 2) = 0$$

$$x^2 + x^2 + 4x + 4 + x - x - 2 = 0$$

$$2x^2 + 4x + 2 = 0 \quad \Rightarrow \quad x^2 + 2x + 1 = 0$$

$$(x + 1)^2 = 0 \quad \Rightarrow \quad x = -1$$

$$at \quad x = -1 \quad \Rightarrow \quad y = -(-1 + 2) = -1$$

the point of contact is $(-1, -1)$.

thank you